

Demonstration of Proposed Features

Date: 04 July 2026 **Team ID:** SOLO-CCFD-2026 **Project Name:** Credit Card Fraud Detection using Machine Learning
Maximum Marks: 1 Mark

S.No	Feature Name	Description	Status (Implemented/Partial/Pending)	Demonstrated (Yes/No)	Remarks
1	Data preprocessing & SMOTE balancing	Scales Time/Amount, balances classes on the training set	Implemented	Yes	Confirmed via class-distribution printout
2	Multi-model training & comparison	Trains Logistic Regression, Decision Tree, Random Forest, XGBoost	Implemented	Yes	Compared via Precision/Recall/F1/AUC
3	Best model auto-selection	Automatically selects the top F1-scoring model and saves it	Implemented	Yes	train_model.py saves best_model.pkl
4	Flask prediction web form	Enter transaction features, get instant Fraud/Genuine prediction	Implemented	Yes	Verified locally at 127.0.0.1:5000
5	Prediction confidence score	Displays fraud probability alongside the prediction	Implemented	Yes	Returned in the /predict JSON response
6	IBM Watson ML deployment	Pushes trained model to a hosted, scalable scoring endpoint	Partial	No	Script ready; live deployment pending IBM Cloud credentials

Feature Implementation Summary

Metric	Value
Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	83%